# Lab 2 - Configuring Basic Single-Area OSPFv2

**Answer Sheet**

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## Part 2: Configure and Verify OSPF Routing

### Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

__show ip route ospf ____________________________________________________________ 2

## Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ____S0/0/0_________ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? __129_______ 2

Does R2 show up as an OSPF neighbor on R1? ___Yes_____ 2

Does R2 show up as an OSPF neighbor on R3? ___No_____ 2

What does this information tell you?

___All traffic from R3 to 192.168.2.0/24 will be routed from R1.____________________________ 2

______________________________________________________________________________

______________________________________________________________________________

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

____R2(config)__________________________________________________ 2

____R2(config-router) ______________________________________________________________

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ____S0/0/1________ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

__65. T1 serial link + R2 Gigabit 0/0 LAN link. ____________________________________________ 2

Is R2 listed as an OSPF neighbor to R3? ___Yes____ 2

# Part 5: Change OSPF Metrics

## Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

__To accurately calculate the cost of high-speed links ____________________

2

## Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

__192.168.3.0/24: R1 S0/0/1 + R3 G0/0 =782____________________

2

__192.168.23.0/30: R1 S0/0/1 + R3 S0/0/1 =845____________________

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

___1562. The route contains two serial links, each costs 781____________________

2

## Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

____OSPF will choose the route with the lowest cost ____________________

4

## Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   Router ID determines the designated router and the backup designated router used by OSPF. 4

2. Why is the DR/BDR election process not a concern in this lab?

   This lab only uses point-to-point links, which do not perform DR/BDR election. 4

3. Why would you want to set an OSPF interface to passive?

   It removes unnecessary OSPF routing information on that interface. 4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   OSPF divides a network into multiple areas. 4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0 4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   ~~RIP~~ EIGRP 0